

BRIAN RIVERA, Ph.D.
Curriculum Vitae

Department of Psychology
St. Olaf College, RNS 160A
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EDUCATION

2021 Ph.D., Educational Psychology, Educational Neuroscience Concentration

Department of Educational Studies, University of Alabama, Tuscaloosa, AL
Advisor: Firat Soylu, Ph.D.

Dissertation: *The neurocognitive mechanisms of arithmetic knowledge verification*

2013 B.A., Scientific and Philosophical Studies of the Mind

Franklin & Marshall College, Lancaster, PA

Areas of Expertise and Research Interests

Educational neuroscience, mathematical and numerical cognition, fraction processing, EEG/ERP methods, multivariate pattern analysis (MVPA) and machine-learning decoding of neural signals, cognitive development, functional brain development in children residing in poverty

PROFESSIONAL EXPERIENCE

St. Olaf College, Northfield, MN

2025–present Visiting Assistant Professor, Department of Psychology

University of Nebraska–Lincoln, Lincoln, NE

2023–2025 Assistant Research Professor, Department of Psychology

2021–2023 Postdoctoral Research Associate, Department of Psychology (Advisor: Timothy Nelson, Ph.D.)

University of Massachusetts Amherst, Amherst, MA

2015–2016 Post-baccalaureate Research Assistant, Cognition and Action Lab (Advisor: Rebecca Spencer, Ph.D.)

SUPERVISED UNDERGRADUATE STUDENT RESEARCH

St. Olaf College

2025–2026 Root Lab students: I. Baker, B. Goldman, M. Hendrickson, R. Herrera Zelaya, E. Leitner, A. Nagy, A. Rodriguez, A. Seiberlich, A. Sticka-Jacobs, P. Weiss, G. Woog, S. Zins
CURI Research (Fall 2026): S. Brooks-Robinson, S. Zins
CURI Research (Spring 2026): M. Hendrickson, A. Sticka-Jacobs
CURI Summer Research (2026): K. Casselman, A. Sticka-Jacobs
McNair Summer Research (2026): R. Herrera Zelaya
Independent Research (Psych 398, Fall 2026): E. Leitner

Advising and Mentoring

2025–2026 Informal advising as visiting faculty at St. Olaf College: letters of recommendation for eleven students applying to study abroad programs, the TRIO McNair Scholars Program,

summer research programs, and internships; regular referral of graduate school, post-baccalaureate, and research opportunities; meetings with students on graduate school plans.

TEACHING EXPERIENCE

Course Instructor

St. Olaf College, Department of Psychology

Psych 390 Electrophysiology of Complex Cognition: EEG Methods and Analysis, Fall 2026 (7 students)

Psych 238 Biopsychology, Fall 2026 (two sections, 51 students)

Psych 390 Electrophysiology of Complex Cognition: EEG Methods and Analysis, Spring 2026 (6 students)

Psych 237 Cognitive Psychology, Spring 2026 (two sections, 26 students each)

Psych 290 A Big History of the Brain, January Term 2026 (21 students)

Psych 238 Biopsychology, Fall 2025 (52 students)

University of Alabama, College of Education

Research Methods in Educational Neuroscience, Spring 2021

Intro to Learning Skills & Strategies, Spring 2020

University of Alabama, OLLI – *A History of Information*, Summer 2019

Teaching Assistant

University of Alabama, College of Education – *Intro to Learning Skills & Strategies*, Fall 2019

University of Alabama, College of Education – *Research Methods in Educational Neuroscience*, Spring 2018

Courses Developed

Psych 390 Electrophysiology of Complex Cognition: EEG Methods and Analysis. New seminar in EEG/ERP methods: student-led sessions on the ERP components and ten MATLAB/EEGLAB labs that build a complete analysis pipeline, with final projects on the ERP CORE open dataset. First offered Spring 2026.

Psych 290 A Big History of the Brain. New January-term course on the evolution of nervous systems and the brain and the history of the ideas used to explain them; counts toward the History and Neuroscience concentrations. First offered January 2026.

Curriculum and Teaching Materials

2026 Open course materials for the EEG/ERP methods seminar (syllabus, lab handouts, and analysis code), documented with St. Olaf Faculty Life Committee support.

2026 Interactive and animated teaching pages on the ERP analysis pipeline for the EEG seminar, published at brivera21.github.io.

2026 Coded animations of the action potential and the voltage-gated sodium channel for Biopsychology.

2025 *ERP Peak Detective: Introduction to Event-Related Potential Analysis with MATLAB*. Peer-reviewed teaching activity in the Teaching Computation in the Sciences Using MATLAB collection, Science Education Resource Center (SERC), Carleton College; rated exemplary.

Prepared to Teach

Introductory Psychology, Biopsychology, Cognitive Psychology, Cognitive Neuroscience, EEG/ERP Methods, Research Methods, Statistics, Learning and Cognition

PEER-REVIEWED PUBLICATIONS (*DENOTES STUDENT CO-AUTHORS)

1. Salehzadeh, R., **Rivera, B.**, Man, K., Jalili, N., & Soylu, F. (2023). EEG decoding of finger numeral configurations with machine learning. *Journal of Numerical Cognition*, 9(1), 206–221. <https://doi.org/10.5964/jnc.10441>
2. **Rivera, B.**, & Soylu, F. (2021). Incongruity in fraction verification elicits N270 and P300 ERP effects. *Neuropsychologia*, 161, 108015. <https://doi.org/10.1016/j.neuropsychologia.2021.108015>
3. Soylu, F., **Rivera, B.**, Anchan, M., & Nathaniel, N. (2019). ERP differences in processing canonical and non-canonical finger numeral representations. *Neuroscience Letters*. <https://doi.org/10.1016/j.neulet.2019.04.032>

Manuscripts Under Review or in Revision

1. Rico-Pico, J., **Rivera, B.**, Fox, N. A., Noble, K. G., & Troller-Renfree, S. V. (under review). Associations between functional brain activity and cognitive skill among children residing in poverty.

Manuscripts in Preparation

1. **Rivera, B.**, & Salehzadeh, R. (in preparation). Multivariate pattern analysis of fraction processing: Dissociating holistic magnitude from component representations in EEG.
2. **Rivera, B.**, Hendrickson, M.*, & Sticka-Jacobs, A.* (in preparation). EEG responses to incorrect arithmetic answers: Evidence of format-dependent arithmetic processing.
3. **Rivera, B.**, Casselman, K.*, Herrera Zelaya, R.*, & Sticka-Jacobs, A.* (in preparation). Graph reasoning and symbolic processing: An ERP study of algebraic reasoning.

Conference Proceedings

1. **Rivera, B.**, & Soylu, F. (2018). Semantic processing in fraction comparison: An ERP study. Paper accepted to the 40th Annual Conference of the Cognitive Science Society. Madison, WI: Cognitive Science Society.

POSTER PRESENTATIONS (*DENOTES STUDENT PRESENTER/CO-AUTHOR)

1. **Rivera, B.** (2026, November). An open-data undergraduate EEG/ERP methods course at a primarily undergraduate institution: Design and first offering. Late-breaking poster accepted for presentation at Neuroscience 2026, Society for Neuroscience, Washington, DC.
2. Casselman, K.*, Herrera Zelaya, R.*, Sticka-Jacobs, A.*, & **Rivera, B.** (2026). Graph Reasoning and Symbolic Processing (GRASP): An ERP study of algebraic reasoning. Poster presented at the UCLA National McNair Conference, Los Angeles, CA.
3. Hendrickson, M.*, Sticka-Jacobs, A.*, & **Rivera, B.** (2026). *EEG responses to incorrect arithmetic answers: Evidence of format-dependent arithmetic processing*. Poster presented at the Minnesota Undergraduate Psychology Conference, Augsburg University, Minneapolis, MN.
4. **Rivera, B.**, & Troller-Renfree, S. (2024). *Associations between functional brain development and cognitive abilities in children born into poverty*. Poster presented at the 2024 Annual Meeting of the Fetal, Infant & Toddler Neuroimaging Group, Baltimore, MD.
5. **Rivera, B.** (2024). *What fraction am I thinking of? Decoding fraction magnitude from EEGs*. Poster presented at the 2024 Annual Meeting of the Mathematical Cognition and Learning Society, Washington, D.C.
6. **Rivera, B.** (2022). *The neurocognitive mechanisms of arithmetic verification in children*. Poster presented at the Cognitive Development Society Biennial Conference, Madison, WI.
7. **Rivera, B.**, & Soylu, F. (2019). *The ERP effects of shared components in fraction comparisons*. Poster presented at the Annual Meeting of the Mathematical Cognition and Learning Society, Carleton University, Ottawa, Canada.

8. **Rivera, B., & Soylu, F. (2018).** *The effects of shared components in fraction comparisons: Evidence from ERPs.* Poster presented at the 6th Biennial International Mind, Brain and Education Conference, University of Southern California, Los Angeles, CA.
9. **Rivera, B. (2018).** *Online experiments with Wolfram technologies.* Poster presented at the 2018 Wolfram Technology Conference, Urbana-Champaign, IL.
10. **Rivera, B., Shannon, N., & Soylu, F. (2017).** *ERP markers for number gesture processing.* Poster presented at SACNAS: The National Diversity in STEM Conference, Salt Lake City, UT.
11. **Rivera, B., & Park, J. (2016).** *Chunking during child visuomotor sequence learning.* SPINES Summer Program, Marine Biological Laboratory, Woods Hole, MA.

PODIUM SESSIONS, COLLOQUIA, AND INVITED PRESENTATIONS

1. **Rivera, B. (2026, March).** *Parts and wholes: Decoding fraction processing insights from EEG brain signals.* Mathematics, Statistics, and Computer Science (MSCS) Department Colloquium, St. Olaf College, Northfield, MN.
2. **Rivera, B. (2024).** *Baby's First Years: An overview of the study and its findings.* Invited talk presented at Science Thursdays, Boys Town National Research Hospital, Omaha, NE.
3. **Rivera, B. (2024).** *Math development and learning difficulties.* Invited guest lecture in Developmental Psychopathology, Teachers College, Columbia University, New York, NY.
4. **Rivera, B. (2022).** *How does the brain do math? Lessons from the study of fractions.* Mathematics Department Colloquium Series, University of Nebraska, Omaha, NE.

GRANTS AND RESEARCH SUPPORT

- 2026 St. Olaf College Professional Development Grant, Faculty Life Committee (\$2,100). Developing and Evaluating an Undergraduate EEG/ERP Methods Course as a Replicable Model for Liberal Arts Institutions (conference travel and registration).
- 2026 St. Olaf College Collaborative Undergraduate Research and Inquiry (CURI) awards supporting undergraduate researchers, spring, summer, and fall 2026.

HONORS AND AWARDS

- 2025 Exemplary rating in peer review of ERP Peak Detective, Teaching Computation in the Sciences collection, Science Education Resource Center (SERC), Carleton College
- 2024 FIT'NG 2024 Young Investigator Award, Fetal, Infant & Toddler Neuroimaging Group, Baltimore, MD
- 2023 ERP Boot Camp Attendance Award, San Diego, CA
- 2018 Wolfram Summer School Attendance Award, Waltham, MA

SERVICE

Departmental and College Service

- 2025–2026 Set up the Department of Psychology's EEG system from existing Biopac equipment, validated the recording protocol, and wrote the laboratory and IRB protocols (approved November 2025) now used by student researchers, St. Olaf College.
- 2025–2026 Trained the student members of the Root Lab in EEG methods and research ethics (CITI certification), St. Olaf College.
- 2026 Faculty mentor, TRIO McNair Scholars Program, St. Olaf College (summer).
- 2026 Participant, Department of Psychology spring student poster session, St. Olaf College.

Reviewing

- 2026 Ad hoc reviewer, *Behavioral and Brain Functions*

PROFESSIONAL MEMBERSHIPS

Society for Neuroscience; Mathematical Cognition and Learning Society

PROFESSIONAL DEVELOPMENT

2025 Participant, Teaching Computation with MATLAB (and GenAI) Workshop, Science Education Resource Center (SERC), Carleton College, Northfield, MN, October 26–28, 2025.

2023 ERP Boot Camp, University of California, San Diego, CA

TECHNICAL SKILLS

EEG and ERP: data collection with BrainVision, Biopac, and EGI systems; preprocessing and analysis in EEGLAB and ERPLAB; multivariate pattern analysis and support-vector decoding of single-trial EEG; ERP CORE

Programming and software: MATLAB, including Psychtoolbox and Live Scripts; Python, including PsychoPy for experiment presentation and scripted figure generation; Mathematica and the Wolfram Language

Languages: Spanish (native) and English